Homework 2

Solutions

1. (Properties of covariance)

   (a) By the definition of covariance and linearity of expectation

$$\begin{aligned}\mathrm{Cov}[\beta\tilde{a}+\alpha,\tilde{b}] &:= \mathrm{E}\left[(\beta\tilde{a}+\alpha-\mathrm{E}[\beta\tilde{a}+\alpha])(\tilde{b}-\mathrm{E}[\tilde{b}])\right] &(1)\\ &= \beta\mathrm{E}\left[(\tilde{a}-\mathrm{E}[\tilde{a}])(\tilde{b}-\mathrm{E}[\tilde{b}])\right] &(2)\\ &= \beta\mathrm{Cov}[\tilde{a},\tilde{b}]. &(3)\end{aligned}$$

   (b) Since $-1 \leq \rho_{\tilde{a},\tilde{b}} \leq 1$ and $\rho_{\tilde{a},\tilde{b}} = \frac{\mathrm{Cov}[\tilde{a},\tilde{b}]}{\sqrt{\mathrm{Var}[\tilde{a}]\mathrm{Var}[\tilde{b}]}}$, it follows that

$$-\sqrt{\mathrm{Var}[\tilde{a}]\mathrm{Var}[\tilde{b}]} \leq \mathrm{Cov}[\tilde{a},\tilde{b}] \leq \sqrt{\mathrm{Var}[\tilde{a}]\mathrm{Var}[\tilde{b}]}. \quad (4)$$

   (c) By the properties in Theorem 8.34,

$$\begin{aligned}\mathrm{Cov}[\tilde{a}+\tilde{b},\tilde{a}-\tilde{b}] &= \mathrm{Cov}[\tilde{a},\tilde{a}-\tilde{b}]+\mathrm{Cov}[\tilde{b},\tilde{a}-\tilde{b}] &(5)\\ &= \mathrm{Cov}[\tilde{a},\tilde{a}]-\mathrm{Cov}[\tilde{a},\tilde{b}]+\mathrm{Cov}[\tilde{b},\tilde{a}]-\mathrm{Cov}[\tilde{b},\tilde{b}] &(6)\\ &= \mathrm{Var}[\tilde{a}]-\mathrm{Var}[\tilde{b}], &(7)\end{aligned}$$

   because $\mathrm{Cov}\left[\tilde{a},\tilde{a}\right] = \mathrm{E}\left[(\tilde{a}-\mathrm{E}\left[\tilde{a}\right])^2\right] = \mathrm{Var}[\tilde{a}]$.

2. (Properties of the sample correlation coefficient)

   (a)

$$\begin{aligned}m(S_X) &= \frac{1}{n}\sum_{i=1}^{n} s(x_i) &(8)\\ &= \frac{1}{n}\sum_{i=1}^{n}\frac{x_i-m(X)}{\sqrt{v(X)}} &(9)\\ &= \frac{1}{\sqrt{v(X)}}\left(\frac{1}{n}\sum_{i=1}^{n}x_i-\frac{1}{n}\sum_{i=1}^{n}m(X)\right) &(10)\\ &= \frac{m(X)-m(X)}{\sqrt{v(X)}} = 0. &(11)\end{aligned}$$

By the same argument, $m(Y) = 0$.

$$v(S_X) = \frac{1}{n-1}\sum_{i=1}^{n}\left(s(x_i) - m(S_X)\right)^2 \tag{12}$$

$$= \frac{1}{n-1}\sum_{i=1}^{n} s(x_i)^2 \tag{13}$$

$$= \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{x_i - m(X)}{\sqrt{v(X)}}\right)^2 \tag{14}$$

$$= \frac{1}{v(X)}\frac{1}{n-1}\sum_{i=1}^{n}\left(x_i - m(X)\right)^2 \tag{15}$$

$$= 1. \tag{16}$$

By the same argument, $v(Y) = 1$.

$$c(S_X, S_Y) = \frac{1}{n-1}\sum_{i=1}^{n}(s(x_i) - m(S_X))(s(x_i) - m(S_Y)) \tag{17}$$

$$= \frac{1}{n-1}\sum_{i=1}^{n} s(x_i)s(y_i) \tag{18}$$

$$= \frac{1}{n-1}\sum_{i=1}^{n}\frac{\left(x_i - m(X)\right)\left(y_i - m(Y)\right)}{\sqrt{v(X)v(Y)}} \tag{19}$$

$$= \rho_{X,Y}. \tag{20}$$

(b) Using the results from part (a) and the formula for the OLS estimator,

$$\frac{1}{n-1}\sum_{i=1}^{n} r_i^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(y_i - \sqrt{v(Y)}\rho_{X,Y}\left(\frac{x - m(X)}{\sqrt{v(X)}}\right) - m(Y)\right)^2 \tag{21}$$

$$= \frac{v(Y)}{n-1}\sum_{i=1}^{n}\left(s(y_i) - \rho_{X,Y}s(x_i)\right)^2 \tag{22}$$

$$= \frac{v(Y)}{n-1}\left(\sum_{i=1}^{n} s(y_i)^2 + \rho_{X,Y}\sum_{i=1}^{n} s(x_i)^2 - 2\rho_{X,Y}\sum_{i=1}^{n} s(x_i)s(y_i)\right) \tag{23}$$

$$= v(Y)\left(1 - \rho_{X,Y}^2\right). \tag{24}$$

(c) By the previous answer,

$$\left(1 - \rho_{X,Y}^2\right)v(Y) \geq \frac{1}{n-1}\sum_{i=1}^{n} r_i^2 \geq 0, \tag{25}$$

which implies that $\rho_{X,Y}^2 \leq 1$, unless $v(Y)$ is zero.

(d) By (b) if

$$\rho_{X,Y} = \pm 1 \tag{26}$$

then

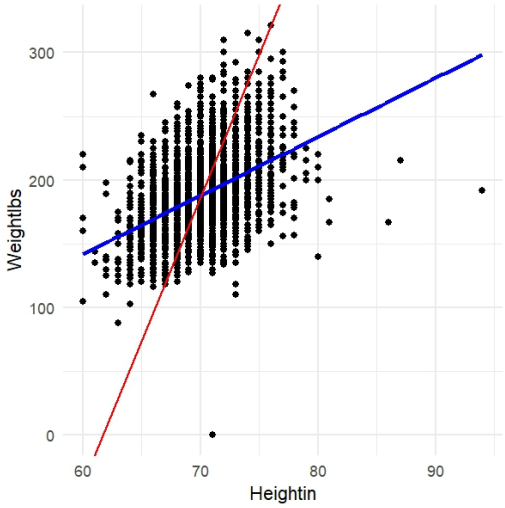
$$\frac{1}{n-1}\sum_{i=1}^{n} r_i^2 = v(Y)\left(1-\rho_{X,Y}^2\right) = 0, \tag{27}$$

which implies that $r_i = 0$ for $1 \leq i \leq n$, so that

$$y_i = \ell_{\text{OLS}}(x_i) \tag{28}$$

$$= \sqrt{v(Y)}\rho_{X,Y}\left(\frac{x-m(X)}{\sqrt{v(X)}}\right) + m(Y). \tag{29}$$

3. (Height and weight)



(a)

(b) The covariance between X and R is 0.000.

(c) ...

(d) Weight variance 896.516, residual variance 712.615, OLS variance 183.901. The sample variance of weights equals the sum of the sample variance of residual and OLS estimates.

(e) R2 0.205, squared correlation between weight and height 0.205.

(f) ...